ADVANCED CONTROL SYSTEMS

Manipulator Dynamics

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Outline

Complements

Linearization

Operational Space Dynamic Model

PROJECT

Complements

Assumption: only the spinning rotor velocity is taken into account in the rotational part of the kinetic energy.

This means that the kinetic energy of the i-th motor (usually located on the link $i-1$) is given by

$$\begin{aligned} T_{m_i} = \frac{1}{2} I_{m_i} \dot{\theta}_{m_i}^2 &= \frac{1}{2} I_{m_i} n_i^2 \dot{q}_i^2 && [n_i \text{ is the gear ratio}] \\ &= \frac{1}{2} B_{m_i} \dot{q}_i^2 && [B_{m_i} > 0] \end{aligned}$$

The total kinetic energy for the n rotors is

$$T_m(\dot{q}) = \sum_{i=1}^{n} T_{m_i}(\dot{q}_i) = \sum_{i=1}^{n} \frac{1}{2} B_{m_i} \dot{q}_i^2 = \frac{1}{2} \dot{q}^T B_m \dot{q}, \qquad \text{with } B_m := \begin{bmatrix} B_{m_1} & 0 & & \\ 0 & B_{m_2} & \ddots & \\ & \ddots & \ddots & 0 \\ & & 0 & B_{m_n} \end{bmatrix} \in \mathbb{R}^{n \times n}$$

with $B_m \succ 0$ by construction and constant.

The dynamic model

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F_v\dot{q} + F_s\text{sign}(\dot{q}) + g(q) = \tau - J^T(q)h_e$$

can be easily integrated with the contribution of B_m as

$$(B(q) + B_m)\,\ddot{q} + C(q,\dot{q})\dot{q} + F_v\dot{q} + F_s\text{sign}(\dot{q}) + g(q) = \tau - J^T(q)h_e$$

since the rotor kinetic energy does not play any role in $C(q,\dot{q})$ because B_m is constant.



τ_i is the torque *after* the gear box of the i-th motor.

$$\tau_i = n_i\tau_{m_i}$$

Bounds on dynamic terms

The matrices in the dynamic model

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F_v\dot{q} + F_s\text{sign}(\dot{q}) + g(q) = \tau - J^T(q)h_e$$

satisfy the following properties

- the element $B_{nn}(q)$ is always constant: $B_{nn}(q) = b_{nn}$
- $k_0 \leq \|B(q)\| \leq k_1 + k_2\|q\| + k_3\|q\|^2$
- $\|C(q,\dot{q})\| \leq (k_4 + k_5\|q\|)\,\|\dot{q}\|$
- $\|g(q)\| \leq k_6 + k_7\|q\|$

If the robot has only revolute joints or bounded prismatic joints ($q_i = d_i \in [d_{i,min}, d_{i,max}]$)

- $k_0 \leq \|B(q)\| \leq k_1$
- $\|C(q,\dot{q})\| \leq k_4\|\dot{q}\|$
- $\|g(q)\| \leq k_6$

Elastic joints

Why *elasticity*?

- Motion transmissions by belts, cables, harmonic drives, etc have *intrinsic* flexibility
- In collaborative robotics / physical Human-Robot interaction the compliance of the robot is increase by inserting *elastic elements* (e.g. Serial Elastic Actuators SEA)

  $\rightarrow$ increase safety

  $\rightarrow$ increase energy efficiency

Why *joint* elasticity?

- flexibility is modeled as *concentrated at the joints* to make the analysis easier
- flexibility is constrained to small deformation (i.e. *linear elastic regime*)

  $\rightarrow$ a stiffness coefficient k_i for the i-th joint

Elastic joints

With elastic elements at each joint, *2n generalized coordinates* are needed to model the manipulator dynamics:

- n before the elastic elements, i.e. at the motors' side after the gear box $\theta \in \mathbb{R}^n$ (i.e. $\theta_i = \frac{\theta_{m,i}}{n_i}$)
- n after the elastic elements, i.e. at the links' side $q \in \mathbb{R}^n$

It is necessary to add the *elastic potential energy* U_e

Let $U_{e,i}$ be the elastic potential energy of the i-th joint

$$U_{e,i} = \frac{1}{2} k_i (q_i - \theta_i)^2 = \frac{1}{2} k_i \left(q_i - \frac{\theta_{m,i}}{n_i} \right)^2, \qquad k_i > 0$$

then ($K \succ 0$)

$$U_e = \sum_{i=1}^{n} U_{e,i} = \sum_{i=1}^{n} \frac{1}{2} k_i (q_i - \theta_i)^2 = \frac{1}{2} (q - \theta)^T K (q - \theta), \text{ where } K := \begin{bmatrix} k_1 & 0 & & \\ 0 & k_2 & \ddots & \\ & \ddots & \ddots & 0 \\ & & 0 & k_n \end{bmatrix} \in \mathbb{R}^{n \times n}$$

The dynamic model obtained solving the Euler equations with the motor kinematic energy (i.e. the inertia matrix B_m) and the elastic potential energy (i.e. the stiffness matrix K) is

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) + K(q-\theta) = 0$$

$$B_m\ddot{\theta} + K(\theta - q) = \tau$$

This is a system of $2n$ second-order differential equations on (q, θ).

If it is necessary to take into account external torques performing work on q and friction effects, the previous equations become

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F\dot{q} + g(q) + K(q-\theta) = -J^T(q)h_e$$

$$B_m\ddot{\theta} + F_m\dot{\theta} + K(\theta - q) = \tau$$

Linearization

Linearization at an equilibrium point q_e

The Lagrangian model

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = u$$

can be written in the *state-space representation* $\dot{x}(t) = f(x,u)$ by defining the state vector x as

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \triangleq \begin{bmatrix} q \\ \dot{q} \end{bmatrix},$$

and re-arranging the implicit differential equations as

$$\begin{aligned} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} x_2 \\ -B^{-1}(x_1)\left(C(x_1,x_2)x_2 + g(x_1)\right) \end{bmatrix} + \begin{bmatrix} 0 \\ B^{-1}(x_1)u \end{bmatrix} \\ &= f(x_1, x_2, u) \end{aligned}$$

We want to derive the *linear dynamic approximation* of the robot which is valid around a given equilibrium point q_e.

Linearization at an equilibrium point q_e

Since q_e is constant, then $\dot{q}_e = 0$. The equilibrium state vector is

$$x_e = \begin{bmatrix} x_{e1} \\ x_{e2} \end{bmatrix} = \begin{bmatrix} q_e \\ 0 \end{bmatrix},$$

Moreover, $(q, \dot{q}) = (q_e, 0) \Rightarrow \ddot{q}_e = 0$.

The nonlinear model evaluated in $(q_e, 0)$

$$B(q_e)\,0 + C(q_e, 0)\,0 + g(q_e) = u_e$$

gives the corresponding 'equilibrium' command

$$u_e = g(q_e)$$

Linearization at an equilibrium point q_e

We now consider the variations around the equilibrium point $(q_e, 0)$ and the command u_e

$$\begin{aligned} q &= q_e + \delta_q \\ \dot{q} &= \dot{q}_e + \dot{\delta}_q = \dot{\delta}_q \\ \ddot{q} &= \ddot{q}_e + \ddot{\delta}_q = \ddot{\delta}_q \\ u &= u_e + \delta_u \end{aligned}$$

we have (around the equilibrium point)

$$\begin{aligned} B(q)\ddot{q} &\simeq B(q_e)\ddot{q}_e + \left.\frac{\partial (B(q)\ddot{q})}{\partial q}\right|_{\substack{q=q_e \\ \dot{q}=0 \\ \ddot{q}=0 \\ u=u_e}} \delta_q + \left.\frac{\partial (B(q)\ddot{q})}{\partial \dot{q}}\right|_{\substack{q=q_e \\ \dot{q}=0 \\ \ddot{q}=0 \\ u=u_e}} \dot{\delta}_q + \left.\frac{\partial (B(q)\ddot{q})}{\partial \ddot{q}}\right|_{\substack{q=q_e \\ \dot{q}=0 \\ \ddot{q}=0 \\ u=u_e}} \ddot{\delta}_q \\ &\simeq B(q_e)\ddot{\delta}_q \end{aligned}$$

Linearization at an equilibrium point q_e

$$
\begin{aligned}
C(q,\dot{q})\dot{q} &\simeq C(q_e,\dot{q}_e)\dot{q}_e + \left.\frac{\partial (C(q,\dot{q})\dot{q})}{\partial q}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \delta_q + \left.\frac{\partial (C(q,\dot{q})\dot{q})}{\partial \dot{q}}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \dot{\delta}_q + \left.\frac{\partial (C(q,\dot{q})\dot{q})}{\partial \ddot{q}}\right|_{\substack{q\\ \dot{q}\\ \ddot{q}\\ u}} \\
&\simeq 0 + o(\delta_q, \dot{\delta}_q)
\end{aligned}
$$

since $\|C(q,\dot{q})\| \leq (k_4 + k_5\|q\|)\,\|\dot{q}\|$, i.e. $(C(q,\dot{q})\dot{q})$ is quadratic w.r.t. $\dot{q}$.
$o(\delta_q, \dot{\delta}_q)$ contains second or higher order infinitesimal terms.

Linearization at an equilibrium point q_e

$$
\begin{aligned}
C(q,\dot{q})\dot{q} &\simeq C(q_e,\dot{q}_e)\dot{q}_e + \left.\frac{\partial(C(q,\dot{q})\dot{q})}{\partial q}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \delta_q + \left.\frac{\partial(C(q,\dot{q})\dot{q})}{\partial \dot{q}}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \dot{\delta}_q + \left.\frac{\partial(C(q,\dot{q})\dot{q})}{\partial \ddot{q}}\right|_{\substack{q\\ \dot{q}\\ \ddot{q}\\ u}} \\
&\simeq 0 + o(\delta_q, \dot{\delta}_q)
\end{aligned}
$$

since $\|C(q,\dot{q})\| \leq (k_4 + k_5\|q\|)\,\|\dot{q}\|$, i.e. $(C(q,\dot{q})\dot{q})$ is quadratic w.r.t. $\dot{q}$.
$o(\delta_q, \dot{\delta}_q)$ contains second or higher order infinitesimal terms.

$$
\begin{aligned}
g(q) &\simeq g(q_e) + \left.\frac{\partial g(q)}{\partial q}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \delta_q + \left.\frac{\partial g(q)}{\partial \dot{q}}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \dot{\delta}_q + \left.\frac{\partial g(q)}{\partial \ddot{q}}\right|_{\substack{q=q_e\\ \dot{q}=0\\ \ddot{q}=0\\ u=u_e}} \ddot{\delta}_q \\
&\simeq g(q_e) + G(q_e)\delta_q
\end{aligned}
$$

where $G(q)$ is the Jacobian matrix $\frac{\partial g(q)}{\partial q}$. (Remark $\|g(q)\| \leq k_6 + k_7\|q\|$)

Linearization at an equilibrium point q_e

The overall ODE around the equilibrium point is

$$B(q_e)\ddot{\delta}_q + g(q_e) + G(q_e)\delta_q + o(\delta_q, \dot{\delta}_q) = u_e + \delta_u$$

and finally

$$B(q_e)\ddot{\delta}_q + G(q_e)\delta_q = \delta_u$$

Let $\delta_x = \begin{bmatrix} \delta_q \\ \dot{\delta}_q \end{bmatrix}$ be the state vector. The state space model around the equilibrium point is

$$\dot{\delta}_x = \underbrace{\begin{bmatrix} 0 & I \\ -B^{-1}(q_e)G(q_e) & 0 \end{bmatrix}}_{\text{'}A\text{' matrix}} \delta_x + \underbrace{\begin{bmatrix} 0 \\ B^{-1}(q_e) \end{bmatrix}}_{\text{'}B\text{' matrix}} \delta_u$$

The matrices 'A' and 'B' are constant.

Linearization along a trajectory $q_d(t)$

It makes more sense to linearize the model around an *equilibrium trajectory* $q_d(t)$.

From $q_d(t)$, we can compute $\dot{q}_d(t)$ and $\ddot{q}_d(t)$ and so the *nominal command* $u_d(t)$

$$B(q_d)\ddot{q}_d + C(q_d, \dot{q}_d)\dot{q}_d + g(q_d) = u_d$$

We now consider the variations around the nominal trajectory

$$\begin{aligned} q(t) &= q_d(t) + \delta_q(t) \\ \dot{q}(t) &= \dot{q}_d(t) + \dot{\delta}_q(t) \\ \ddot{q}(t) &= \ddot{q}_d(t) + \ddot{\delta}_q(t) \\ u(t) &= u_d(t) + \delta_u(t) \end{aligned}$$

i.e.

$$B(q_d + \delta_q)(\ddot{q}_d + \ddot{\delta}_q) + C(q_d + \delta_q, \dot{q}_d + \dot{\delta}_q)(\dot{q}_d + \dot{\delta}_q) + g(q_d + \delta_q) = u_d + \delta_u$$

Linearization along a trajectory $q_d(t)$

Let's now compute the first order approximation.

Inertia matrix

$$B(q_d + \delta_q) \simeq B(q_d) + \sum_{i=1}^{n} \left. \frac{\partial B_i(q)}{\partial q} \right|_{q=q_d} e_i^T \delta_q$$

Coriolis and centrifugal term

$$C(q_d + \delta_q, \dot{q}_d + \dot{\delta}_q)(\dot{q}_d + \dot{\delta}_q) = c(q_d + \delta_q, \dot{q}_d + \dot{\delta}_q) \simeq c(q_d, \dot{q}_d) + \left. \frac{\partial c(q, \dot{q})}{\partial q} \right|_{\substack{q=q_d \\ \dot{q}=q_d}} \delta_q + \left. \frac{\partial c(q, \dot{q})}{\partial \dot{q}} \right|_{\substack{q=q_d \\ \dot{q}=q_d}} \dot{\delta}_q$$

and so

$$C(q_d + \delta_q, \dot{q}_d + \dot{\delta}_q)(\dot{q}_d + \dot{\delta}_q) \simeq C(q_d, \dot{q}_d)\dot{q}_d + C_q(q_d, \dot{q}_d)\delta_q + C_{\dot{q}}(q_d, \dot{q}_d)\dot{\delta}_q$$

Gravity term

$$g(q_d + \delta_q) \simeq g(q_d) + G(q_d)\delta_q$$

Linearization along a trajectory $q_d(t)$

Cancelling out the terms related to the nominal trajectory and the higher order terms

$$\left(B(q_d) + \sum_{i=1}^{n} \left.\frac{\partial B_i(q_d)}{\partial q}\right|_{q=q_d} e_i^T \delta_q\right)(\ddot{q}_d + \ddot{\delta}_q) + C(q_d, \dot{q}_d)\dot{q}_d + C_q(q_d, \dot{q}_d)\delta_q + C_{\dot{q}}(q_d, \dot{q}_d)\dot{\delta}_q + \\ + g(q_d) + G(q_d)\delta_q = u_d + \delta_u$$

we end up with the following dynamic model on the variations

$$B(q_d)\ddot{\delta}_q + \sum_{i=1}^{n} \left.\frac{\partial B_i(q_d)}{\partial q}\right|_{q=q_d} \ddot{q}_d e_i^T \delta_q + C_q(q_d, \dot{q}_d)\delta_q + C_{\dot{q}}(q_d, \dot{q}_d)\dot{\delta}_q + G(q_d)\delta_q = \delta_u$$

By defining the "spring"-like term

$$D(q_d, \dot{q}_d, \ddot{q}_d) \triangleq \sum_{i=1}^{n} \left.\frac{\partial B_i(q_d)}{\partial q}\right|_{q=q_d} \ddot{q}_d e_i^T + C_q(q_d, \dot{q}_d) + G(q_d)$$

Linearization along a trajectory $q_d(t)$

We finally get

$$B(q_d)\ddot{\delta}_q + C_{\dot{q}}(q_d, \dot{q}_d)\dot{\delta}_q + D(q_d, \dot{q}_d, \ddot{q}_d)\delta_q = \delta_u$$

The equivalent state-space model around the nominal trajectory is

$$\dot{\delta}_x = \underbrace{\begin{bmatrix} 0 & I \\ -B^{-1}(q_d)D(q_d, \dot{q}_d, \ddot{q}_d) & -B^{-1}(q_d)C_{\dot{q}}(q_d, \dot{q}_d) \end{bmatrix}}_{\text{`}A\text{' matrix}} \delta_x + \underbrace{\begin{bmatrix} 0 \\ B^{-1}(q_d) \end{bmatrix}}_{\text{`}B\text{' matrix}} \delta_u$$

The matrices '$A(t)$' and '$B(t)$' are *time-varying*.

We have a time-varying linear system.

Operational Space Dynamic Model

We end up with the following model for the manipulator having the generalized torque as input (τ) and the generalized coordinates as output ($\dot{q}, \ddot{q}$)

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau - J^T(q)h_e$$

How does this model change if we use the Cartesian coordinate $x = \begin{bmatrix} p \\ \phi \end{bmatrix}$ of the end-effector?

The determination of the dynamic model with Lagrange formulation using operational space variables allows a complete description of the system motion only in the case of a *nonredundant manipulator*, i.e.

$$\text{size}(x) = \text{size}(q),$$

otherwise internal motions could occur.

Operational Space Dynamic Model

Torques at the end effector h_e corresponding to joint torques τ_e

$$\tau_e = J^T(q)h_e$$

Relationship between q and x: Direct kinematics

$$x = \kappa(q)$$

Relationship between $\dot{q}$ and $\dot{x}$: Analytical Jacobian

$$\dot{x} = J_A(q)\dot{q}, \qquad J_A(q) := \frac{\partial \kappa(q)}{\partial q} = \begin{bmatrix} J_P(q) \\ J_\phi(q) \end{bmatrix} = \begin{bmatrix} \frac{\partial p}{\partial q} \\ \frac{\partial \phi}{\partial q} \end{bmatrix}$$

$$J = T_A(\phi)J_A, \qquad T_A(\phi) = \begin{bmatrix} I & 0 \\ 0 & T(\phi) \end{bmatrix}, \qquad \omega = T(\phi)\dot{\phi}$$

where ω is the angular velocity with respect to the base frame, while $\dot{\phi}$ are the nonorthogonal components of angular velocity defined with respect to the axes of a frame that varies as the end-effector orientation varies.
Relationship between $\ddot{q}$ and $\ddot{x}$

$$\ddot{x} = J_A(q)\ddot{q} + \dot{J}_A(q,\dot{q})\dot{q}$$

We end up with

$$B_A(x)\ddot{x} + C_A(x,\dot{x})\dot{x} + g_A(x) = u - u_e$$

where

$$\begin{aligned} B_A(x) &\overset{(*)}{=} (J_A B^{-1} J_A^T)^{-1} \\ C_A(x,\dot{x})\dot{x} &= B_A J_A B^{-1} C\dot{q} - B_A \dot{J}_A \dot{q} \\ g_A(x) &= B_A J_A B^{-1} g \\ u &= T_A^T(x) h \\ u_e &= T_A^T(x) h_e \end{aligned}$$

$(*)$ Assumption: B_A nonsingular, i.e. J_A full rank (no kinematic and representation singularities)

For non-redundant manipulator in a nonsingular configuration, we have the simplified expression

$$\begin{aligned} B_A(x) &= J_A^{-T} B J_A^{-1} \\ C_A\dot{x} &= J_A^{-T} C\dot{q} - B_A \dot{J}_A \dot{q} \\ g_A(x) &= J_A^{-T} g \\ u &= T_A^T(x) h \\ u_e &= T_A^T(x) h_e \end{aligned}$$

$(*)$ Assumption: B_A nonsingular, i.e. J_A full rank (no kinematic and representation singularities)


Project

PROJECT – Assignment # 5

To do

- Compute the dynamic model in the operational space